

FitBuddy – AI Fitness Plan Generator using Gemini Models

Project Description:

FitBuddy is a web-based application that uses AI to generate personalized workout plans and nutrition tips based on a user's fitness goals, such as weight loss, muscle gain, or general wellness. Built with FastAPI and integrated with Google's Gemini AI models, FitBuddy simplifies the process of creating structured and effective fitness routines.

The application takes in user information—like age, weight, goal, and preferred workout intensity—and generates a customized 7-day fitness plan. Users can also submit feedback to update and refine their plan using AI. A complementary nutrition or recovery tip is also provided based on their selected goal.

With a clean user interface and database support via SQLite, FitBuddy ensures each user's data and generated plan are stored and accessible. It offers both API endpoints and an interactive HTML interface for ease of use.

Scenarios:

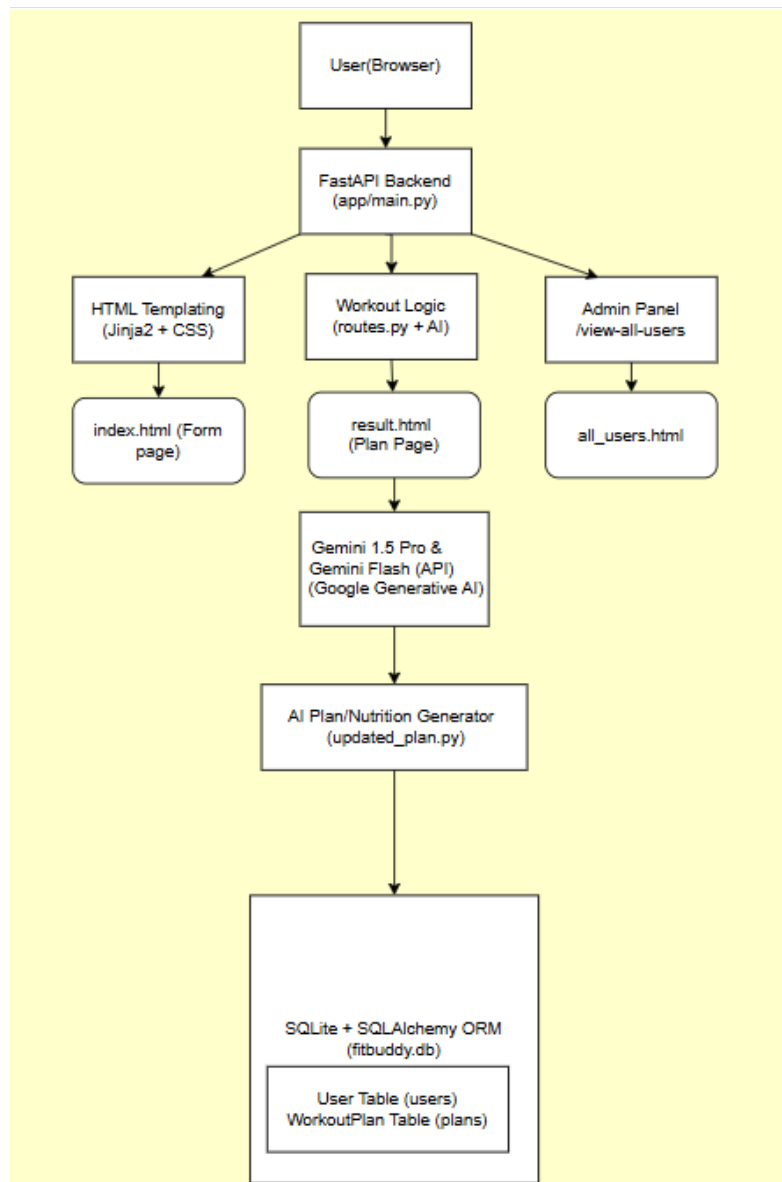
Scenario 1 : A user inputs their personal details including name, age, weight, fitness goal (e.g., “weight loss” or “muscle gain”), and preferred workout intensity (“high”, “medium” or “low”) to receive a personalized 7-day workout plan. The system generates a day-by-day schedule tailored to the user's fitness objective and experience level. This AI-driven customization enhances user motivation and adherence by providing structured and goal-specific routines.

Scenario 2: A user submits feedback on their existing workout plan, such as suggesting changes like “more focus on cardio” or “include more rest days.” The system uses this input to regenerate an updated workout plan, refining it based on the user's evolving needs. This feedback loop ensures that users stay engaged and receive dynamic fitness support as their preferences or progress change.

Scenario 3: A user requests a nutrition or recovery tip based on their selected fitness goal. The platform delivers a concise and relevant suggestion (e.g., “Include protein in your post-workout meal” for muscle gain), which complements the workout plan. These targeted tips help users improve results by aligning their diet and recovery practices with their physical training regimen.

Scenario 4: An admin or coach accesses the platform to view all registered users and their workout plans through a dedicated dashboard. This view allows them to track individual progress, compare original and updated plans, and monitor user activity. It streamlines oversight for trainers managing multiple clients or students.

Technical Architecture:



Pre - requisites:

1. FastAPI Framework Knowledge: [FastAPI Documentation](#)
2. Gemini API Familiarity: [Google Generative AI Documentation](#)
3. HTML, CSS, and Jinja2 Template Skills: [W3Schools HTML/CSS/Jinja2 Tutorials](#)
4. Python Programming Proficiency: [Python Official Docs](#)
5. Version Control with Git: [Git Documentation](#)
6. SQLAlchemy and SQLite Basics: [SQLAlchemy Docs](#)
7. Environment Setup (pip, virtualenv, or conda): [Virtualenv Guide](#)
8. Uvicorn ASGI Server: [Uvicorn Docs](#)

Project Workflow:

Milestone 1. Model selection and Architecture

- Activity 1.1. Research and Select the Appropriate Generative AI Model
- Activity 1.2. Define the architecture of the application
- Activity 1.3. Set up the development Environment
- Milestone 2. Core functionalities Development
 - Activity 2.1. Develop the core functionalities
 - Activity 2.2. Implement the FastAPI Backend to Manage Routing and User Input Processing
- Milestone 3. route.py Development
 - Activity 3.1. Writing the Main Application Logic in routes.py
- Milestone 4. Frontend Development
 - Activity 4.1. Designing and Developing User Interface
 - Activity 4.2. Creating Dynamic Templates with FastAPI's Jinja2
- Milestone 5. Deployment
 - Activity 5.1. Preparing the Application for Local Deployment
 - Activity 5.2. Testing and Verifying Local Deployment

Milestone 1. Model selection and Architecture

In this milestone, we focus on selecting the appropriate **generative AI models from Google's Gemini family** for the **FitBuddy** application. This includes evaluating the performance and capabilities of multiple Gemini models to ensure they align with the application's core functionalities—such as generating personalized workout plans, updating them based on feedback, and providing tailored nutrition or recovery advice.

Activity 1.1: Research and Select the Appropriate Generative AI Model

Understand the Project Requirements

FitBuddy's AI model needed to:

- Generate structured 7-day workout plans
- Provide concise and actionable nutrition tips
- Update workout plans based on user feedback
- Perform well in real-time, web-based interaction

Model Evaluation

We initially experimented with several Hugging Face models:

- T5 & BART – For text generation (workout/nutrition)
- DistilBERT – For potential intent classification
- Bloom – For general purpose generation

While these models showed promise in basic text generation, they had notable limitations.

These models required: